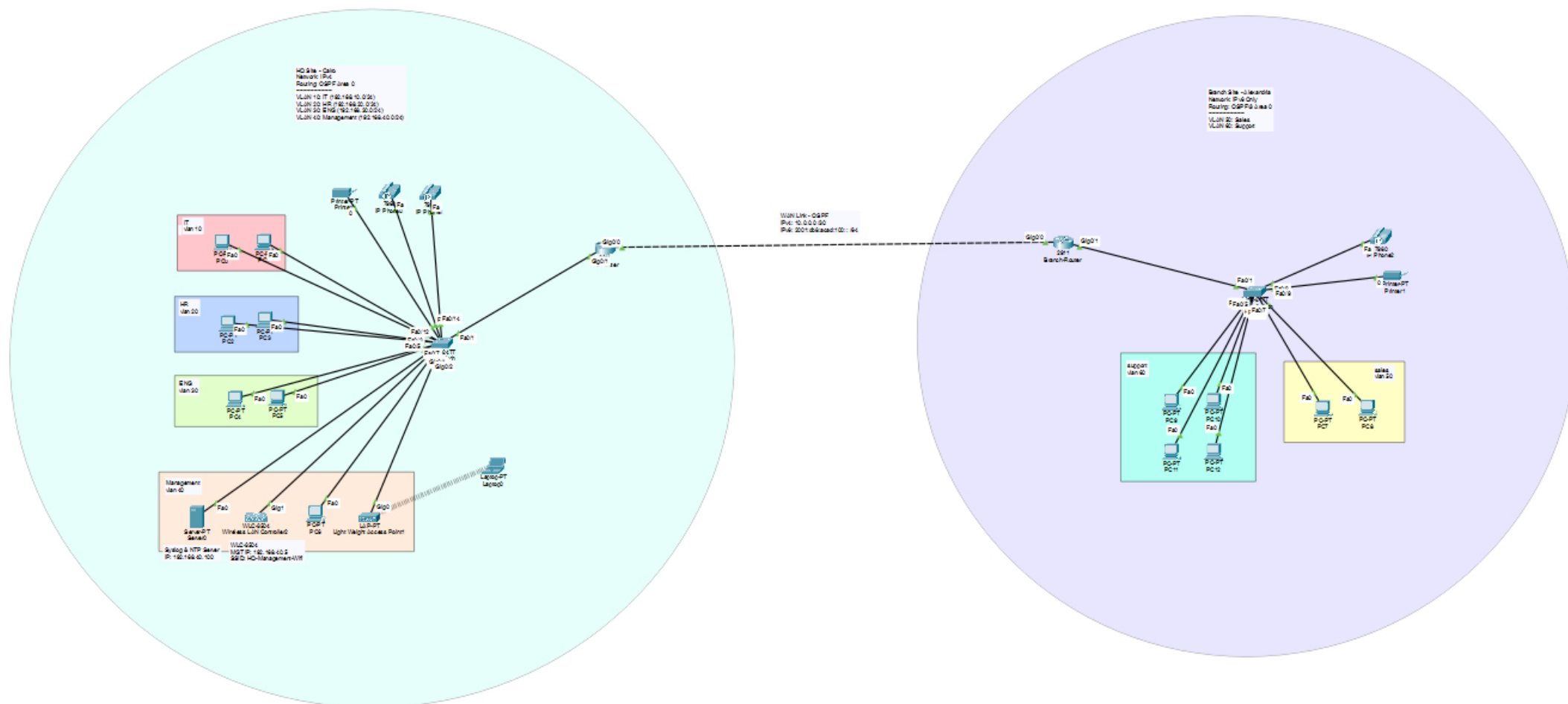




WAN Link - OSPF  
IPv4: 10.0.0.0/30  
IPv6: 2001:db8:acad:100::/64

Branch Site - Alexandria  
Network: IPv6 Only  
Routing: OSPFv3 Area 0  
VLAN 50: Sales  
VLAN 60: Support

Gig0/0  
2911  
Branch-Router  
Gig0/1

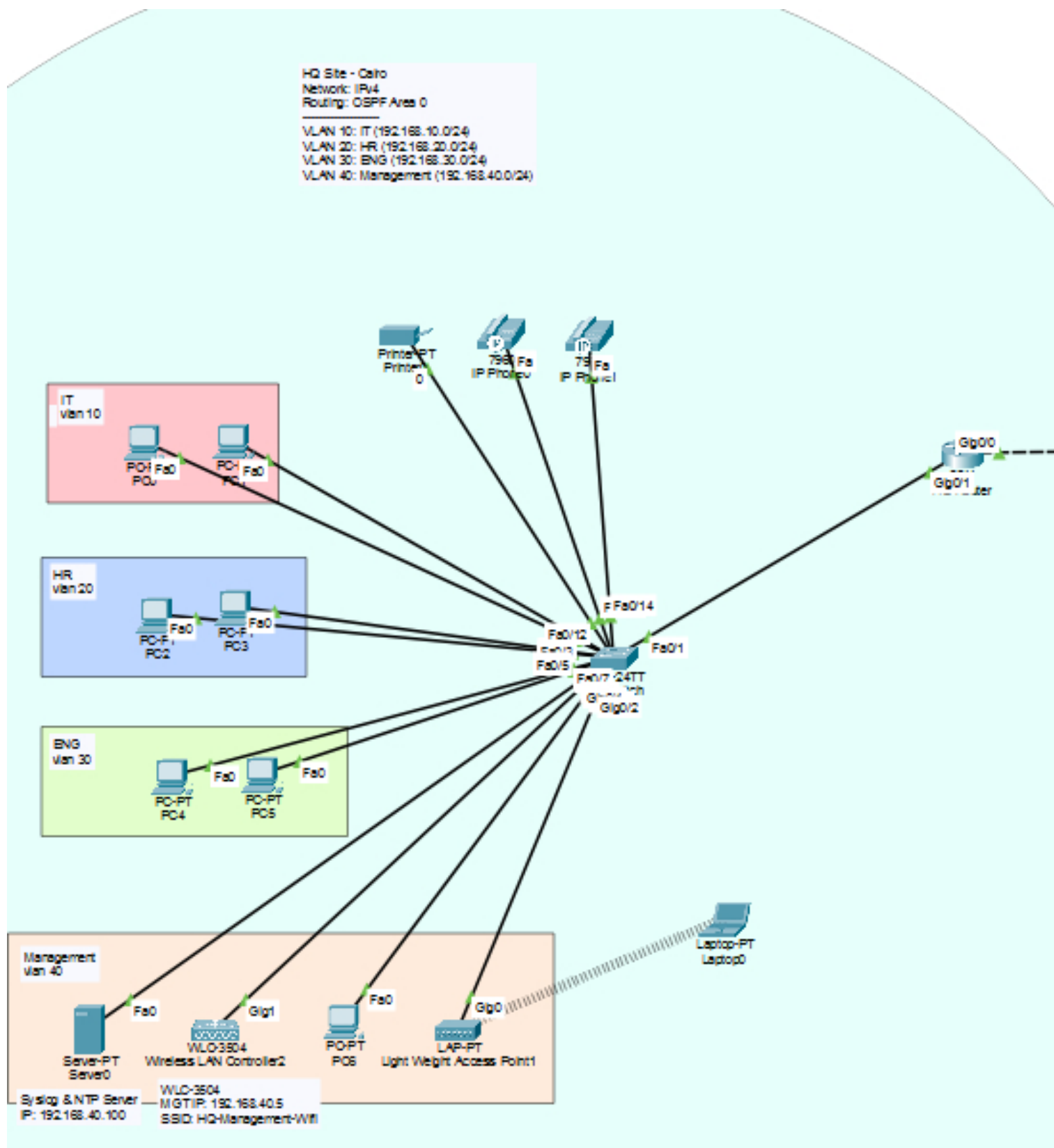
IP  
Fa 7960  
Phone2  
0  
Printer-PT  
Printer1

support  
vlan 60  
PC-PT  
PC9  
Fa0  
PC-PT  
PC11  
Fa0  
PC-PT  
PC10  
Fa0  
PC-PT  
PC12  
Fa0

sales  
vlan 50  
Fa0  
PC-PT  
PC7  
Fa0  
PC-PT  
PC8

HQ Site - Cairo  
Network: IPv4  
Routing: OSPF Area 0

VLAN 10: IT (192.168.10.0/24)  
VLAN 20: HR (192.168.20.0/24)  
VLAN 30: ENG (192.168.30.0/24)  
VLAN 40: Management (192.168.40.0/24)



Physical Config CLI Attributes

## IOS Command Line Interface

```
HQ-Router#
HQ-Router#sh
HQ-Router#show ip ro
HQ-Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1.10
  192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1.20
  192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/1.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/1.30
  192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/1.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/1.40

HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
```

Copy

Paste

Physical Config CLI Attributes

## IOS Command Line Interface

```
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
O  2001:DB8:ACAD:50::/64 [110/2]
   via FE80::2E0:F9FF:FEB4:1601, GigabitEthernet0/0
O  2001:DB8:ACAD:60::/64 [110/2]
   via FE80::2E0:F9FF:FEB4:1601, GigabitEthernet0/0
C  2001:DB8:ACAD:100::/64 [0/0]
   via GigabitEthernet0/0, directly connected
L  2001:DB8:ACAD:100::1/128 [0/0]
   via GigabitEthernet0/0, receive
L  FF00::/8 [0/0]
   via Null0, receive
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#
```

Copy

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## IOS Command Line Interface

```
Branch-Router#
Branch-Router#sh
Branch-Router#show ipv
Branch-Router#show ipv6 r
Branch-Router#show ipv6 rou
Branch-Router#show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:ACAD:50::/64 [0/0]
    via GigabitEthernet0/1.50, directly connected
L   2001:DB8:ACAD:50::1/128 [0/0]
    via GigabitEthernet0/1.50, receive
C   2001:DB8:ACAD:60::/64 [0/0]
    via GigabitEthernet0/1.60, directly connected
L   2001:DB8:ACAD:60::1/128 [0/0]
    via GigabitEthernet0/1.60, receive
C   2001:DB8:ACAD:100::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L   2001:DB8:ACAD:100::2/128 [0/0]
    via GigabitEthernet0/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
Branch-Router#
Branch-Router#
Branch-Router#
Branch-Router#
```

Copy

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HQ-Switch

PhysicalConfigCLIAttributes

IOS Command Line Interface

HQ-Switch#  
HQ-Switch#sh  
HQ-Switch#show vl  
HQ-Switch#show vlan b  
HQ-Switch#show vlan brief

| VLAN | Name               | Status | Ports                                                                                                                       |
|------|--------------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/9, Fa0/11, Fa0/12, Fa0/13<br>Fa0/14, Fa0/15, Fa0/16, Fa0/17<br>Fa0/18, Fa0/19, Fa0/20, Fa0/21<br>Fa0/22, Fa0/23, Fa0/24 |
| 10   | IT                 | active | Fa0/2, Fa0/3                                                                                                                |
| 20   | HR                 | active | Fa0/4, Fa0/5                                                                                                                |
| 30   | ENG                | active | Fa0/6, Fa0/7                                                                                                                |
| 40   | Management         | active | Fa0/8, Fa0/10, Gig0/1, Gig0/2                                                                                               |
| 1002 | fddi-default       | active |                                                                                                                             |
| 1003 | token-ring-default | active |                                                                                                                             |
| 1004 | fddinet-default    | active |                                                                                                                             |
| 1005 | trnet-default      | active |                                                                                                                             |

HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#  
HQ-Switch#

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☐ Top

Branch-Switch

PhysicalConfigCLIAttributes

IOS Command Line Interface

Branch-Switch>  
Branch-Switch>  
Branch-Switch>sh  
Branch-Switch>show v  
Branch-Switch>show vl  
Branch-Switch>show vlan b  
Branch-Switch>show vlan brief

| VLAN | Name               | Status | Ports                                                                                                                                                        |
|------|--------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | default            | active | Fa0/8, Fa0/9, Fa0/10, Fa0/11<br>Fa0/12, Fa0/13, Fa0/14, Fa0/15<br>Fa0/16, Fa0/17, Fa0/18, Fa0/19<br>Fa0/20, Fa0/21, Fa0/22, Fa0/23<br>Fa0/24, Gig0/1, Gig0/2 |
| 50   | Sales              | active | Fa0/6, Fa0/7                                                                                                                                                 |
| 60   | Support            | active | Fa0/2, Fa0/3, Fa0/4, Fa0/5                                                                                                                                   |
| 1002 | fddi-default       | active |                                                                                                                                                              |
| 1003 | token-ring-default | active |                                                                                                                                                              |
| 1004 | fddinet-default    | active |                                                                                                                                                              |
| 1005 | trnet-default      | active |                                                                                                                                                              |

Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>  
Branch-Switch>

CopyPaste

☐ Top

HQ-Router

Physical Config CLI Attributes

IOS Command Line Interface

```
HQ-Router#
HQ-Router#
HQ-Router#
HQ-Router#sh
HQ-Router#show ru
HQ-Router#show running-config
Building configuration...

Current configuration : 2226 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname HQ-Router
!
!
!
!
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp excluded-address 192.168.40.1 192.168.40.100
!
ip dhcp pool IT_POOL
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
ip dhcp pool HR_POOL
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
ip dhcp pool ENG_POOL
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
ip dhcp pool MGT_POOL
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
option 43 ip 192.168.40.5
!
!
!
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
username admin privilege 15 secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
license udi pid CISCO2911/K9 sn FTX1524GHSB-
!
!
```

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

**SYSLOG**

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

Syslog

Syslog

Service

OnOff

|    | Time                       | HostName     | Message                                                                   |
|----|----------------------------|--------------|---------------------------------------------------------------------------|
| 1  | -                          | 192.168.40.1 | ...                                                                       |
| 2  | -                          | 192.168.40.1 | %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.40.100 port 514 ... |
| 3  | -                          | 10.0.0.2     | ...                                                                       |
| 4  | -                          | 10.0.0.2     | %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.40.100 port 514 ... |
| 5  | -                          | 192.168.40.1 | ...                                                                       |
| 6  | -                          | 192.168.40.1 | ...                                                                       |
| 7  | -                          | 192.168.40.1 | ...                                                                       |
| 8  | -                          | 192.168.40.1 | ...                                                                       |
| 9  | -                          | 192.168.40.1 | ...                                                                       |
| 10 | -                          | 192.168.40.1 | ...                                                                       |
| 11 | -                          | 192.168.40.1 | ...                                                                       |
| 12 | -                          | 192.168.40.1 | ...                                                                       |
| 13 | -                          | 192.168.40.1 | ...                                                                       |
| 14 | -                          | 192.168.40.1 | ...                                                                       |
| 15 | -                          | 192.168.40.1 | ...                                                                       |
| 16 | 09.23.2026 09:45:44.319 PM | 192.168.40.1 | ...                                                                       |
| 17 | 09.23.2026 09:45:44.319 PM | 192.168.40.1 | ...                                                                       |
| 18 | 09.23.2026 09:45:44.319 PM | 192.168.40.1 | ...                                                                       |
| 19 | 09.23.2026 09:45:44.319 PM | 192.168.40.1 | ...                                                                       |

Clear Log

Top

HQ-Router

HQ-Cairo(config)#interface loopback 1

HQ-Cairo(config-if)#shutdown

HQ-Cairo(config-if)#no shutdown

HQ-Cairo(config-if)#exit

\*Sep 23, 21:45:44.4545: %LINK-5-CHANGED: Interface Loopback1, changed state to administratively down

\*Sep 23, 21:45:44.4545: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to down

\*Sep 23, 21:45:44.4545: %LINK-3-UPDOWN: Interface Loopback1, changed state to down

\*Sep 23, 21:45:44.4545: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up

HQ-Cairo(config)#

HQ-Cairo(config)#

HQ-Cairo(config)#

Copy

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PC2

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceFastEthernet0

IP Configuration

☒ DHCP☐ Static

IPv4 Address192.168.20.2

Subnet Mask255.255.255.0

Default Gateway192.168.20.1

DNS Server0.0.0.0

IPv6 Configuration

☐ Automatic☒ Static

IPv6 Address/

Link Local AddressFE80::250:FFF:FED4:DD17

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

Username

Password

☐ Top

Laptop0

PhysicalConfigDesktopProgrammingAttributes

IP ConfigurationX

InterfaceWireless0

IP Configuration

☒ DHCP☐ Static

IPv4 Address192.168.40.103

Subnet Mask255.255.255.0

Default Gateway192.168.40.1

DNS Server0.0.0.0

IPv6 Configuration

☒ Automatic☐ Static

IPv6 Address/

Link Local AddressFE80::2D0:D3FF:FE43:22B6

Default Gateway

DNS Server

☐ Top

PC9

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address 2001:DB8:ACAD:60:260:70FF:FE6A:5849 / 64

Link Local Address FE80::260:70FF:FE6A:5849

Default Gateway FE80::2E0:F9FF:FEB4:1602

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC7

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address 2001:DB8:ACAD:50:290:CFF:FEE6:CBC4 / 64

Link Local Address FE80::290:CFF:FEE6:CBC4

Default Gateway FE80::2E0:F9FF:FEB4:1602

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC6

Physical

Config

Desktop

Programming

Attributes

Web Browser

< > URL https://192.168.40.5/frameWlan.html

Go Stop

CISCO

MONITOR

WLANs

CONTROLLER

WIRELESS

SECURITY

MANAGEMENT

COMMANDS

HELP

FEEDBACK

Save Configuration

Ping

Logout

Refresh

Home

WLANs

WLANs

Advanced

AP Groups

WLANs

Entries 1 - 1 of 1

Current Filter:

Change Filter

Clear Filter

Create New Go

| <input type="checkbox"/> | WLAN ID | Type | Profile Name | WLAN SSID | Admin Status | Security Policies |                   |
|--------------------------|---------|------|--------------|-----------|--------------|-------------------|-------------------|
| <input type="checkbox"/> | 1       | WLAN | HQMWIFI      | HQMWIFI   | Enabled      | [WPA2][Auth(PSK)] | <div>Remove</div> |

Top

PC6

PhysicalConfigDesktopProgrammingAttributes

Web Browser

<>URLhttps://192.168.40.5/frameWlanEdit.html

GoStop

Save Configuration | Ping | Logout | Refresh

CISCO

MONITORWLANsCONTROLLERWIRELESSSECURITYMANAGEMENTCOMMANDSHelpFEEDBACK

Home

WLANs

WLANs

Advanced

AP Groups

WLANs > Edit 'HQMWIFI'

< BACK

Apply

GeneralSecurityQoSPolicy-MappingAdvanced

FlexConnect

FlexConnect Local Switching2

☒ Enabled

FlexConnect Local Auth12

☒ Enabled

Learn Client IP Address5

☒ Enabled

Vlan based Central Switching13

☐ Enabled

Central DHCP Processing

☐ Enabled

Override DNS

☐ Enabled

NAT-PAT

☐ Enabled

Central Assoc

☐ Enabled

11k

Assisted Roaming Prediction Optimization

☐ Enabled

Neighbor List

☐ Enabled

Neighbor List Dual Band

☐ Enabled

Radius Client Profiling

DHCP Profiling

☐

HTTP Profiling

☐

Local Client Profiling

DHCP Profiling

☐

HTTP Profiling

☐

Universal AP Admin Support

Universal AP Admin

☐

11v BSS Transition Support

BSS Transition

☐

Disassociation Imminent

☐

Disassociation Timer(0 to 3000 TBTT)

200

Optimized Roaming Disassociation Timer(0 to 40 TBTT)

40

Tunneling

Tunnel Profile

None

mDNS

mDNS Snooping

☐ Enabled

Foot Notes

1 Web Policy cannot be used in combination with IPsec

2(a) FlexConnect Local Switching is not supported with IPsec, CRANITE authentication, Override Interface ACLs

2(b) When flexconnect local authentication is enabled, irrespective of AP on connected or standalone mode the AP will act as NAS

2(c) When flexconnect local authentication is disabled, AP on connected mode will use WLC as NAS and AP as NAS while its on standalone mode

3 When client exclusion is enabled, a Timeout Value of zero means infinity (will require administrative override to reset excluded clients)

☐ Top

Physical Config CLI Attributes

## IOS Command Line Interface

```
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#ex  
HQ-Router#exit
```

HQ-Router con0 is now available

Press RETURN to get started.

Unauthorized Access is Strictly Prohibited!

```
HQ-Router>
```

Copy

Paste

## Command Prompt

```
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#ssh -l admin 192.168.40.1
```

Password:

Unauthorized Access is Strictly Prohibited!

```
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#  
HQ-Router#
```



PC2



Physical

Config

Desktop

Programming

Attributes

Command Prompt



Cisco Packet Tracer PC Command Line 1.0

C:\&gt;ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time&lt;1ms TTL=254

Reply from 10.0.0.2: bytes=32 time&lt;1ms TTL=254

Reply from 10.0.0.2: bytes=32 time&lt;1ms TTL=254

Reply from 10.0.0.2: bytes=32 time&lt;1ms TTL=254

Ping statistics for 10.0.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\&gt;

## Command Prompt



Cisco Packet Tracer PC Command Line 1.0

C:\>

C:\>

C:\>

C:\>ping 192.168.40.100

Pinging 192.168.40.100 with 32 bytes of data:

Reply from 192.168.40.100: bytes=32 time=39ms TTL=128

Reply from 192.168.40.100: bytes=32 time=22ms TTL=128

Reply from 192.168.40.100: bytes=32 time=13ms TTL=128

Reply from 192.168.40.100: bytes=32 time=18ms TTL=128

Ping statistics for 192.168.40.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 13ms, Maximum = 39ms, Average = 23ms

C:\>